

Lab 8 – Web App Pentesting

Q1. Run the apache server

- Sudo systemctl start apache2

```
File Actions Edit View Help
(kali@host-3-57)-[~]
$ sudo systemctl start apache2
(kali@host-3-57)-[~]
$ sudo systemctl reload apache2
(kali@host-3-57)-[~]
$
```

Q2. Add Web page

Sudo nano /var/www/html/hello.html

```
(kali@host-3-57)-[~]
$ cd /var/www/html
(kali@host-3-57)-[/var/www/html]
$ ls
hello.html  index.html  index.nginx-debian.html
(kali@host-3-57)-[/var/www/html]
$ cat hello.html
<html>
<body>
<h1>HOST</h1>
<p>
I am not a hacker!
</p>
</body>
</html>
```

```
(kali@host-3-57)-[/var/www/html]
$ chmod 701 /var/www/html/hello.html
chmod: changing permissions of '/var/www/html/hello.html': Operation not permitted
(kali@host-3-57)-[/var/www/html]
$ chmod o+r /var/www/html/hello.html
chmod: changing permissions of '/var/www/html/hello.html': Operation not permitted
```

Trying to change permissions of the created directory.

Question 2 Add a web page

Lets create a webpage in the main apache document directory, known as the DocumentRoot. Use your favourite editor.

```
sudo nano /var/www/html/hello.html
```

The contents of this file should be:

```
<html>
<body>
<div>HOST</div>
<p>
I am clever
</p>
</body>
</html>
```

The "/var/www/html/hello.html" file must be readable by other. This should be the default. If you create directories in this DocumentRoot by others. However if needed you can always do this manually:

```
chmod 701 /var/www/html/some_directory
chmod o-r /var/www/html/hello.html
```

You can direct your browser to see this page by using the URL

```
http://yourmachinename/hello.html
```

Replace "yourmachinename" with the output of running the "hostname" command, eg

```
[root@host-19-17 kali]# hostname
host-19-17.linuxzoo.net
```

Note that, once working, you can access your linuxzoo web pages just by doing http:// in front of your hostname, or your virtual host URL as normal.

Check Tests: Complete

hello.html is readable by others and owned by root	Pass
hello.html contains the word HOST (case sensitive)	Pass
http://host/hello.html actually works	Pass

Q3. Adding directories

- Create directories – sudo mkdir
- Set permissions – chmod 701
- Copy files – sudo cp text from hello.html → web, vm
- Change text – sudo nano HOST → Web, VM

Question 3 Add two new directories/files

Create the following directories, each of which must be executable for other:

- /var/www/html/web
- /var/www/html/vm

Remember to use sudo

In each of these new directories create a file called "hello.html", identical to hello.html from /var/www/html/public_html, except in "web/hello.html" replace the word HOST with WEB. In "vm/hello.html" replace the word HOST with VM. Case is important.

Check Tests: Complete

Can read http://.web/hello.html	Pass
http://.web/hello.html contains WEB	Pass
Can read http://.vm/hello.html	Pass
http://.web/hello.html contains VM	Pass

```
(kali@ host-3-57) ~$ sudo mkdir /var/www/html/vm
(kali@ host-3-57) ~$ sudo chmod 701 /var/www/html/web
(kali@ host-3-57) ~$ sudo chmod 701 /var/www/html/vm
(kali@ host-3-57) ~$ sudo cp /var/www/html/public_html/hello.html /var/www/html/web
cp: cannot stat '/var/www/html/public_html/hello.html': No such file or directory
(kali@ host-3-57) ~$ sudo find /var/www/ -name hello.html
/var/www/html/hello.html
(kali@ host-3-57) ~$ sudo find /var/www/ -name public_html
(kali@ host-3-57) ~$ sudo cp /var/www/html/hello.html /var/www/html/web
(kali@ host-3-57) ~$ sudo cp /var/www/html/hello.html /var/www/html/vm
(kali@ host-3-57) ~$ sudo nano /var/www/html/web/hello.html
(kali@ host-3-57) ~$ sudo nano /var/www/html/vm/hello.html
(kali@ host-3-57) ~$
```

Q4. Create 2 virtual hosts

Check hostname

-web , vm

Sudo nano

Edit hostname and DocumentRoot

```
File Actions Edit View Help
GNU nano 8.2
<VirtualHost *:80>
  ServerAdmin web-3-57.linuxzoo.net
  DocumentRoot /var/www/html/web
  ServerName www.mywebsite.com
</VirtualHost>

<VirtualHost *:80>
  ServerAdmin vm-3-57.linuxzoo.net
  DocumentRoot /var/www/html/vm
  ServerName www.mywebsite.com
</VirtualHost>
```

```
(kali@host-3-57)-[/]
$ sudo nano /etc/apache2/sites-available/mysite.conf

(kali@host-3-57)-[/]
$
```

Now, activate the files

Sudo a2ensite mysite

Then, reload apache and check syntax OK

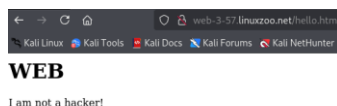
```
(kali@host-3-57)-[/]
$ sudo a2ensite mysite
Enabling site mysite.
To activate the new configuration, you need to run:
  systemctl reload apache2

(kali@host-3-57)-[/]
$ sudo systemctl reload apache2

(kali@host-3-57)-[/]
$ apache2ctl -t
Syntax OK

(kali@host-3-57)-[/]
$
```

Now, visit to the own page from browser



<http://web-3-57.linuxzoo.net/hello.html>

<http://vm-3-57.linuxzoo.net/hello.html>

Q5. Basic Authentication

```
---(kali@host-3-57)-[/]
$ sudo mkdir /var/www/html/tom
---(kali@host-3-57)-[/]
$ sudo chmod 701 /var/www/html/tom
---(kali@host-3-57)-[/]
$ sudo nano /var/www/html/tom > p1.html
zsh: permission denied: p1.html
---(kali@host-3-57)-[/]
$ sudo nano /var/www/html/tom/p1.html
---(kali@host-3-57)-[/]
$ sudo cat /var/www/html/tom/p1.html
<html>
<body>
<h1>TOM</h1>
<p>
Document body goes here.
</p>
</body>
</html>
---(kali@host-3-57)-[/]
$ sudo chmod 701 /var/www/html/tom/p1.html
---(kali@host-3-57)-[/]
$ sudo chmod 644 /var/www/html/tom/p1.html
---(kali@host-3-57)-[/]
$
```

1. Create file mkdir
2. Make it executable and access permission – chmod 701
3. Create another file p1.html and make readable by others

Sudo nano → create

Readable → chmod 644

Question 5 Basic Authentication

These questions are concerned with the configuration of Basic Authentication in apache.

Create a directory in /var/www/html called "tom" and ensure it is executable by others. This should be the default permissions.

Now within that new directory "tom", create a file p1.html and ensure it is readable by others. This should be the default permissions.

```
<html>
<body>
<h1>TOM</h1>
<p>
Document body goes here.
</p>
</body>
</html>
```

Use can if needed use the chmod commands from earlier, except this time on tom and on p1.html rather than hello.html. Again

Check	Tests: Complete
Apache Running	PASS
p1.html is readable by others	PASS
p1.html contains the word TOM (case sensitive)	PASS
http://host/tom/p1.html actually works	PASS

Q6. Basic Auth permission

```
---(kali@host-3-57)-[/]
$ sudo cat /etc/apache2/sites-available/mysite.conf
<VirtualHost *:80>
    ServerAdmin noreply@linuxzoo.net
    DocumentRoot /var/www/html/web
    ServerName web-3-57.linuxzoo.net
</VirtualHost>

<VirtualHost *:80>
    ServerAdmin noreply@linuxzoo.net
    DocumentRoot /var/www/html/vm
    ServerName vm-3-57.linuxzoo.net
</VirtualHost>

<Directory /var/www/html/tom/>
    AllowOverride FileInfo AuthConfig Limit Indexes
    Options MultiViews Indexes SymLinksIfOwnerMatch IncludesNoExec
    Require method GET POST OPTIONS
</Directory>
---(kali@host-3-57)-[/]
```

Reload - sudo systemctl reload apache2

Q7. Add two new directories/files

Done similar steps for these new directories
and give permission

- Replace TOM -> Harry

and TOM -> RICHARD

- Permit to each file “chmod 701”

```
(kali@host-3-49)-[/]
$ sudo cat /var/www/html/tom/harry/p3.html
<html>
<body>
<h1>HARRY</h1>
<p>
Document body goes here.
</p>
</body>
</html>

(kali@host-3-49)-[/]
$ sudo cat /var/www/html/tom/richard/p2.html
<html>
<body>
<h1>RICHARD</h1>
<p>
Document body goes here.
</p>
</body>
</html>
```

```
(kali@host-3-49)-[/]
$ sudo chmod 701 /var/www/html/tom/richard

(kali@host-3-49)-[/]
$ sudo chmod 701 /var/www/html/tom/harry

(kali@host-3-49)-[/]
$ ls -l /var/www/html/tom/richard
ls: cannot open directory '/var/www/html/tom/richard': Permission denied

(kali@host-3-49)-[/]
$ sudo cat /var/www/html/tom/richard/p2.html
<html>
<body>
<h1>RICHARD</h1>
<p>
Document body goes here.
</p>
</body>
</html>

(kali@host-3-49)-[/]
$ sudo cat /var/www/html/tom/harry/p3.html
<html>
<body>
<h1>HARRY</h1>
<p>
Document body goes here.
</p>
</body>
</html>
```

Question 7 Add two new directories/files

Create the following directories in the tom directory, each of which must be executable by others:

- /var/www/html/tom/richard
- /var/www/html/tom/harry

In each of these new directories create a file similar to p1.html, but called:

- /var/www/html/tom/richard/p2.html
- /var/www/html/tom/harry/p3.html

In "richard/p2.html" replace the word TOM with RICHARD. In "harry/p3.html" replace the word TOM with HARRY. Case is important.

Check	Tests: Complete
Can read http://.tom/richard/p2.html	PASS
http://.tom/richard/p2.html contains RICHARD	PASS
Can read http://.tom/harry/p3.html	PASS
http://.tom/harry/p3.html contains HARRY	PASS

Q8. Basic Auth file

Answer: Adding a password to the users :

1. User: richard , Password: pass1

2. User: harry, Password: pass2

```
(kali@host-3-49)-[/]
$ sudo htpasswd -c /var/www/webpasswd richard
New password:
Re-type new password:
Adding password for user richard

(kali@host-3-49)-[/]
$ sudo htpasswd /var/www/webpasswd harry
New password:
Re-type new password:
Adding password for user harry
```

Question 8 Basic Auth file

Create a password file for basic authentication in /var/www/.
Remember to use sudo.

The htpasswd command allows you to create the file, and to add users to the file. Use it to create a basic authentication password file called "/var/www/webpasswd". Put into this file two users with the following passwords:

User: richard Password: pass1
User: harry Password: pass2

Check	Tests: Complete
/var/www/webpasswd exists and seems readable	PASS
Contents semi-sensible for richard?	PASS
Contents semi-sensible for harry?	PASS

Q9. Secure richard/

Answer:

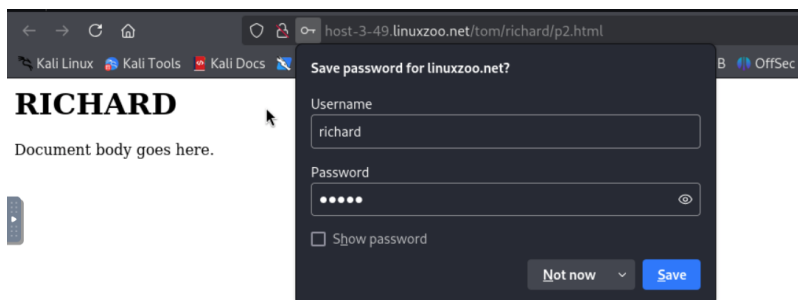
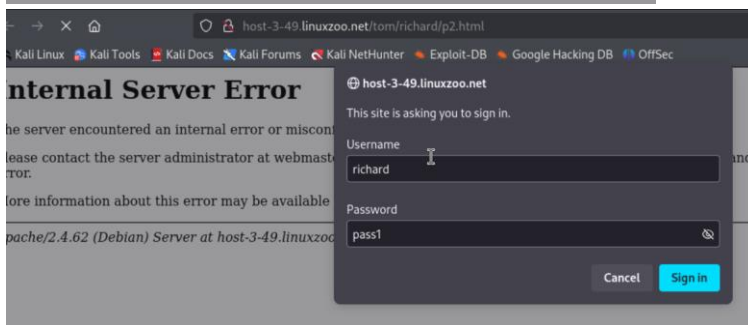
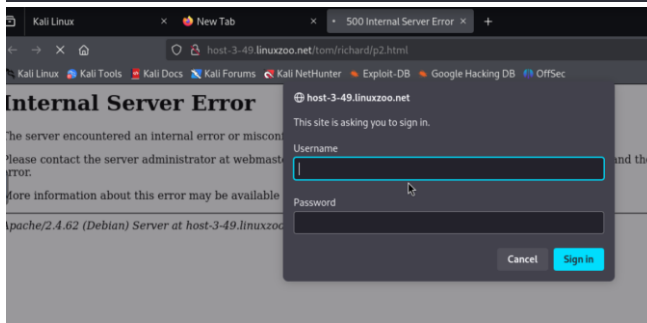
Secure website or webpage , and need grant access while logging in -

User

Pass

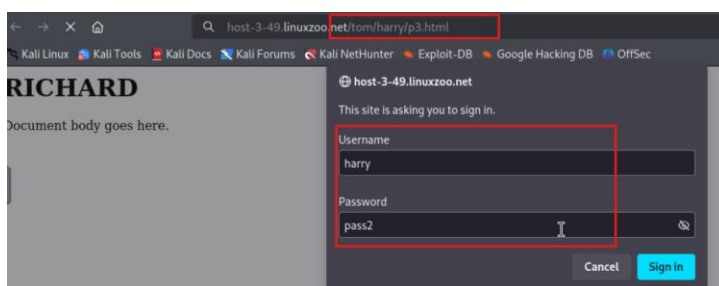
```
(kali@host-3-49)-[/]
$ sudo nano /var/www/html/tom/richard/.htaccess

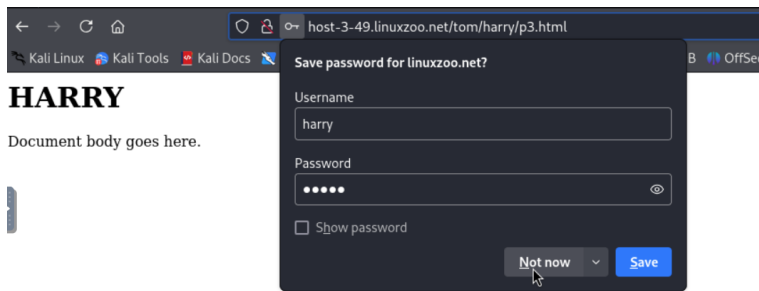
(kali@host-3-49)-[/]
$ sudo cat /var/www/html/tom/richard/.htaccess
AuthType Basic
AuthName "Restricted Area"
AuthUserFile /var/www/webpasswd
Require user richard
```



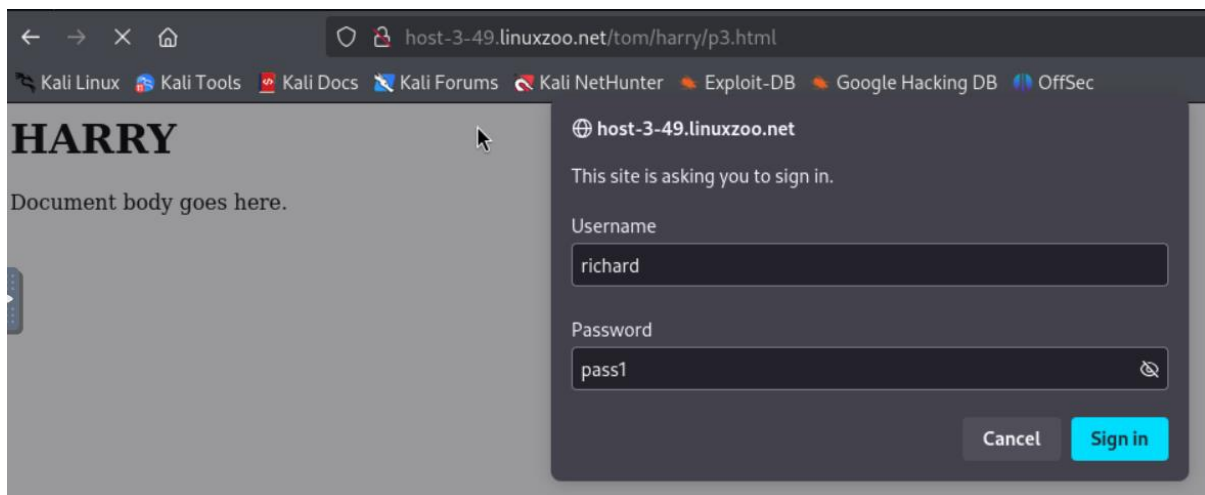
Q10. Secure Harry

Answer: Similar to Richard





Now, We create a group and attach the passwords of harry and richard. Which means while logging to harry or richard – both passwords should work.



```
(kali@host-3-49)-[/]
$ namei -l /var/www/webgroup
f: /var/www/webgroup
drwxr-xr-x root root /
drwxr-xr-x root root var
drwxr-xr-x root root www
-rw-r--r-- www-data www-data webgroup

(kali@host-3-49)-[/]
$ sudo chown root:root /var/www/webgroup

(kali@host-3-49)-[/]
$ sudo chmod 644 /var/www/webgroup

(kali@host-3-49)-[/]
$ sudo systemctl restart apache2

(kali@host-3-49)-[/]
$
```